

Trustworthy Federated Graph Neural Networks: Unifying Fairness, Privacy, Robustness, and Explainability for High-Stakes Risk Detection

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Abstract

Federated Learning (FL) enables institutions to train shared models over sensitive, decentralised graph data—transactions, credit records, social ties—without exchanging raw data. Yet accuracy alone does not make such systems deployable: recent regulatory and scientific frameworks (the EU AI Act, the NIST AI Risk Management Framework) demand that automated decision systems be simultaneously *fair*, *private*, *robust*, and *transparent*. Existing work addresses these properties in isolation, and no prior method combines fairness, differential privacy, and Byzantine robustness for federated graph neural networks (GNNs). We present **FedFairGNN**, a unified trustworthy federated graph-learning framework built from three components: (i) *Fairness-Sensitive Edge Reweighting* (FSER), which suppresses the biased cross-group message passing responsible for structural discrimination; (ii) *Fairness-Targeted Gradient Decomposition* (FTGD), which achieves differential privacy for the sensitive attribute by privatising only the low-dimensional demographic-parity statistics—making privacy nearly free where standard full-gradient DP-SGD destroys utility; and (iii) *Bi-objective Frank–Wolfe Aggregation* (BFWA) and its Byzantine-robust variant, which enforce a hard fairness budget during aggregation while screening malicious clients, including a new fairness-poisoning threat. We further instrument the framework with predictive uncertainty, GNN attention explanations, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF compliance mapping. Across five real datasets spanning credit, recidivism, social, and—uniquely—large-scale crypto (Elliptic Bitcoin, 203,769 nodes) domains, FedFairGNN attains the best fairness–utility–privacy–robustness trade-off among ten baselines, including recent fairness-aware federated and graph methods. All tables and figures are regenerated from logged runs; code and experiments are released for full reproducibility. This work supports SDG 9 by advancing trustworthy, infrastructure-grade collaborative AI.

Keywords: Federated learning; Graph neural networks; Algorithmic fairness; Differential privacy; Byzantine robustness; Explainable AI; Trust metrics; Regulatory compliance.

1 Introduction

High-stakes decisions in finance and public administration—credit approval, fraud and anti-money-laundering (AML) screening, pretrial risk—are increasingly mediated by machine-learning models operating over *relational* data. Graph Neural Networks (GNNs) are the de facto tool for these tasks because fraud, default, and recidivism propagate along relationships between users, accounts, and transactions [12, 20, 30]. In practice the relevant graph is fragmented across institutions—different banks, exchanges, or agencies—each legally barred from pooling raw records under regimes such as the GDPR. Federated Learning (FL) [25] resolves this by training a shared model through the exchange of model updates rather than data.

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However, deployability is no longer defined by accuracy alone. The European Commission’s Ethics Guidelines for Trustworthy AI, the EU AI Act, and the U.S. NIST AI Risk Management Framework all require that automated decision systems be *fair, privacy-preserving, robust, transparent, and accountable*. Federated graph learning strains every one of these axes simultaneously. **Fairness:** GNN message passing amplifies structural bias, so that members of a demographic group connected to risky neighbours are flagged disproportionately, and the server cannot measure this directly because it never sees the data. **Privacy:** model updates leak information about local records, yet the standard remedy—differentially private SGD—injects isotropic noise across every parameter and, on the compact GNNs used here, destroys utility (Section 5.4). **Robustness:** a single malicious or malfunctioning client can corrupt the global model, and we show that fairness mechanisms open a *new* attack surface—a client can report fabricated fairness statistics to capture a fairness-aware aggregator. **Transparency:** relational models are opaque, complicating the audits that regulation demands.

Prior work treats these properties in isolation: fair GNNs assume centralised data [10, 21, 34]; fair FL largely targets tabular or image data [15, 22]; robust aggregation ignores fairness [7, 35]. To our knowledge no method jointly delivers fairness, differential privacy, and Byzantine robustness for federated GNNs, and none has been evaluated on large-scale crypto graphs.

We close this gap with **FedFairGNN**, a unified framework whose contributions are:

1. **Fairness-Sensitive Edge Reweighting (FSER):** a learnable attention correction that identifies and suppresses biased cross-group edges during message passing, mitigating structural discrimination at its source (Section 4.1).
2. **Fairness-Targeted Gradient Decomposition (FTGD):** the observation that the sensitive attribute enters the model *only* through the demographic-parity statistics lets us obtain (ϵ, δ) -differential privacy by privatising two bounded-sensitivity scalars rather than the full gradient. This makes privacy essentially free—where DP-FedAvg collapses utility—and we prove the guarantee via a self-contained Rényi-DP accountant (Section 4.2).
3. **Bi-objective Frank–Wolfe Aggregation (BFWA)** and a **Byzantine-robust variant:** server-side constrained optimisation enforces a hard fairness budget during aggregation, while distance-based screening defends against poisoning—including a novel fairness-poisoning attack we introduce (Sections 4.3, 5.6).
4. **A trustworthiness instrumentation layer:** predictive uncertainty and calibration, GNN attention explanations with fairness attribution, a transparent composite *trust score*, sustainability accounting, and an explicit EU AI Act / NIST RMF compliance mapping with an auto-generated model card (Sections 4.4, 6).
5. **A comprehensive, reproducible evaluation** on five real datasets spanning credit, recidivism, social, and large-scale crypto (Elliptic Bitcoin, 203,769 nodes) domains, against ten baselines including recent fairness-aware federated and graph methods. FedFairGNN attains the best overall trust trade-off; every reported number is regenerated from logged runs.

By advancing trustworthy, infrastructure-grade collaborative AI, this work supports UN Sustainable Development Goal 9. The remainder of the paper reviews related work (Section 2), formalises the setting (Section 3), details the method (Section 4), reports experiments (Section 5), analyses trustworthiness and compliance (Section 6), and discusses limitations (Section 7).

2 Related Work

Fairness in graph neural networks. GNNs inherit and amplify bias through homophilous message passing. Remedies include adversarial debiasing (FairGNN [10]), counterfactual and

stability regularisation (NIFTY [2]), pre-processing that minimises inter-group distributional distance (EDITS [11]), sensitive-information neutralisation via heterogeneous neighbours (FairSIN [34]), re-balancing through counterfactual mixup (FairGB [23]), invariant learning across inferred sensitive partitions (FairINV [37]), and disentangled debiasing (DAB-GNN [21]). All assume *centralised* data. Our FSER is closest in spirit to the edge-reweighting variants of FairSIN and to distribution-aware reweighting [6], but operates inside a federated, private, and robust pipeline.

Fairness in federated learning. Two notions coexist. *Client (performance) fairness* equalises accuracy across clients: q-FedAvg [22] up-weights high-loss clients. *Group fairness* equalises outcomes across demographic groups: FairFed [15] adjusts aggregation weights by each client’s local fairness gap; FedFB ports FairBatch to FL; PUFFLE [9] blends a demographic-parity loss with DP-SGD under a target-disparity controller; FedFACT [3] gives provable global-and-local group fairness. These target tabular or image data. For *graphs*, F²GNN [28] combines a fairness-weighted aggregation with structure-aware group-balance weights. The most recent efforts sit closest to our setting: FairGFL [36] adds DP to federated fair GNNs but targets *cross-client* (performance) fairness arising from imbalanced overlapping subgraphs rather than demographic group fairness; FedGraph-Fair [16] applies distributionally-robust optimisation for personalised group-fair federated graph learning; FaVGNN [31] (2026) achieves group fairness in *vertical* FL via completion-driven adversarial fusion; and FDP-Fair [33] (2026) combines federated DP training with demographic-parity post-processing. We compare against FairFed, q-FedAvg, FedFB, F²GNN, FairSIN (run federated), and —as the strongest 2026 competitors— horizontal adaptations of FaVGNN and FDP-Fair. FairGFL is discussed but not benchmarked head-to-head, as it optimises a different (cross-client) fairness objective.

Differential privacy in FL. DP-FedAvg [26] clips per-client updates and adds Gaussian noise; privacy is tracked with the moments accountant / Rényi DP [1, 29]. On the compact GNNs used for tabular graphs, isotropic gradient noise at meaningful ϵ overwhelms the signal (Section 5.4). FedFDP [24] adapts clipping to fairness but still privatises the whole gradient. FTGD instead privatises only the *sufficient statistics* of the fairness term, exploiting the fact that the sensitive attribute never enters the task loss—a targeted mechanism whose (ϵ, δ) guarantee follows from the Gaussian mechanism and post-processing immunity.

Byzantine-robust aggregation. Krum and Multi-Krum [7], coordinate-wise median and trimmed mean [35], Bulyan [14], and trust-bootstrapping schemes such as FLTrust [8] defend against malicious updates under attacks including sign-flipping, model-replacement [4], ALIE [5], and IPM [32]. None targets fairness. Recently, fairness-poisoning attacks emerged—EAB-FL [27] and the fairness-constrained optimisation attack [19]—that *increase* disparity while preserving accuracy, evading both robust and fairness-aware defences. We reproduce the fairness-constrained optimisation attack of Kasyap et al. [19] and defend with a robust fairness-constrained aggregator.

Explainability, uncertainty, and trust metrics. GNN explainers attribute predictions to edges/features; we use the FSER attention itself as a native, audit-ready explanation and add gradient-based attribution of the *disparity*. Predictive uncertainty via MC-dropout [17] and calibration (ECE, Brier) quantify confidence reliability. Composite trust scoring for FL is nascent; we contribute a transparent power-mean aggregation over utility, fairness, privacy, robustness, and calibration. Finally, we align the system to the EU AI Act and NIST AI RMF, operationalising accountability as an auditable artifact rather than prose.

Positioning. No prior method unifies fairness, differential privacy, and Byzantine robustness for federated GNNs, nor evaluates on large-scale crypto graphs. FedFairGNN does both, and reports the full trustworthiness picture—including uncertainty, explainability, sustainability, and regulatory mapping—under one framework.

3 Preliminaries and Problem Setting

Cross-silo federated graph learning. There are K institutions (clients). Client k holds a private graph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k, \mathbf{y}_k, \mathbf{s}_k)$ with node features $\mathbf{X}_k \in \mathbb{R}^{n_k \times d}$, binary task labels $\mathbf{y}_k \in \{0, 1\}^{n_k}$ (e.g. fraud/default/recidivism), and a binary *sensitive attribute* $\mathbf{s}_k \in \{0, 1\}^{n_k}$ (e.g. gender, age, race). Raw graphs never leave the client. A server coordinates T communication rounds; each round clients perform E local epochs and transmit model parameters, which the server aggregates. We evaluate the global model on the pooled held-out test nodes of all clients. Statistical heterogeneity is modelled by a symmetric Dirichlet(α) split over labels, the standard FL non-IID model [18]; smaller α is more skewed.

Group fairness. For predicted probabilities $\hat{y}$ and sensitive attribute s we report three group-fairness metrics (lower is fairer):

$$\text{DPD} = |\mathbb{E}[\hat{y} \mid s=0] - \mathbb{E}[\hat{y} \mid s=1]| \quad (\text{Demographic Parity Difference}), \quad (1)$$

$$\text{EOD} = |\text{TPR}_{s=0} - \text{TPR}_{s=1}| \quad (\text{Equal Opportunity Difference}), \quad (2)$$

$$\Delta_{\text{EOdds}} = \max(|\Delta \text{TPR}|, |\Delta \text{FPR}|) \quad (\text{Equalized Odds}). \quad (3)$$

DPD is computed on soft scores (matching the differentiable training term); EOD and Equalized Odds use thresholded predictions.

Differential privacy. A randomised mechanism $\mathcal{M}$ is (ϵ, δ) -differentially private if for all adjacent datasets D, D' and all outputs S , $\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$ [13]. We protect the *sensitive attribute*: adjacency is defined by flipping one node’s s_i . The Gaussian mechanism adds $\mathcal{N}(0, \sigma^2)$ noise calibrated to the L_2 sensitivity of the released quantity; composition across rounds is tracked with Rényi DP [29]. For the Gaussian mechanism with noise multiplier $z = \sigma/\Delta$, the Rényi divergence of order α is $\epsilon_{\text{RDP}}(\alpha) = \alpha/(2z^2)$; over T releases it composes as $T\alpha/(2z^2)$, and the tight conversion to (ϵ, δ) -DP is $\epsilon = \min_{\alpha>1} [\epsilon_{\text{RDP}}(\alpha) + \log(1/\delta)/(\alpha - 1)]$.

Threat model for robustness. Up to $f < K/2$ clients are Byzantine and may send arbitrary updates. We consider data poisoning (label flipping), classical model poisoning (Gaussian, sign-flip, scaling/model-replacement [4]), the strong omniscient attacks ALIE [5] and IPM [32], and a *fairness-poisoning* attack: a malicious client transmits a biased, scaled update while reporting fabricated fairness/utility statistics ($\text{DPD} \approx 0$, high AUC) to capture a fairness-aware aggregator. The server is honest but curious and never accesses raw data.

Objective. We seek a global model that jointly (i) maximises task utility (AUC-ROC), (ii) minimises group unfairness, (iii) satisfies a target (ϵ, δ) -DP for the sensitive attribute, and (iv) retains utility and fairness under Byzantine corruption—the trustworthiness trade-off that motivates Section 4.

4 The FedFairGNN Framework

FedFairGNN augments federated GNN training with three components acting at complementary points—inside message passing (FSER), at the local gradient (FTGD), and at server aggregation

(BFWA)—plus a trustworthiness instrumentation layer. Figure-free overview: each round, the server broadcasts θ_{t-1} ; each client runs E FSER-GNN local epochs with FTGD to obtain a privatised update; the server aggregates updates under a fairness budget with BFWA (optionally Byzantine-robust) to obtain θ_t .

4.1 Fairness-Sensitive Edge Reweighting (FSER)

In a homophilous graph, nodes of sensitive group A structurally tied to positive (risky) neighbours are flagged even absent genuine signal. FSER adds a learnable correction to graph attention. For edge (i, j) with head embeddings $\mathbf{h}_i, \mathbf{h}_j$, the standard attention logit $e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{h}_i \parallel \mathbf{h}_j])$ is corrected by a *fairness risk score*

$$\phi_{ij} = \mathbb{I}(s_i \neq s_j) \cdot \max(0, \cos(\mathbf{h}_i, \mathbf{h}_j)), \quad \tilde{e}_{ij} = e_{ij} - \beta^{(l)} \phi_{ij}, \quad (4)$$

so that *cross-group* edges between similarly-embedded nodes—the pattern most responsible for propagating demographic bias—are down-weighted after the per-node softmax. The suppression strength $\beta^{(l)} \geq 0$ is learned per layer (clamped to $[0, 5]$). Because ϕ_{ij} and the resulting attention are retained, FSER is also *self-explaining*: Section ?? audits how much attention mass FSER removes from cross-group edges.

4.2 Fairness-Targeted Gradient Decomposition (FTGD)

The local objective is $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{fair}}$, with weighted BCE task loss and the differentiable soft demographic-parity penalty $\mathcal{L}_{\text{fair}} = |\mu_0 - \mu_1|$, where $\mu_g = \frac{1}{n_g} \sum_{i:s_i=g} \hat{y}_i$ is the group- g mean prediction.

Key observation. The sensitive attribute s enters the entire update *only* through the two scalars (μ_0, μ_1) ; the task loss is a function of $(\mathbf{X}, \mathbf{y})$ alone. Therefore, to achieve DP with respect to s , it suffices to privatise (μ_0, μ_1) —two bounded-sensitivity scalars—rather than the full $|\theta|$ -dimensional gradient.

Mechanism. Flipping one node’s group membership changes (μ_0, μ_1) by at most $\Delta = \sqrt{1/n_0^2 + 1/n_1^2}$ in L_2 . We release $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$ with noise multiplier z , form $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$, and backpropagate. Additionally, to reduce task/fairness gradient conflict we orthogonalise the task gradient against the fairness gradient (gradient surgery) before the step.

Proposition 1 (Privacy of FTGD). *Each local FTGD epoch is $(\alpha, \alpha/(2z^2))$ -RDP with respect to the sensitive attribute. Over TE epochs the mechanism is $(\alpha, TE \alpha/(2z^2))$ -RDP, and hence $(\min_{\alpha>1} [TE \alpha/(2z^2) + \log(1/\delta)/(\alpha - 1)], \delta)$ -DP.*

Proof sketch. Releasing $(\tilde{\mu}_0, \tilde{\mu}_1)$ is the Gaussian mechanism at sensitivity Δ and noise $z\Delta$, giving noise multiplier z and RDP $\alpha/(2z^2)$ [29]. Every subsequent quantity—the fairness gradient, the model update, all future predictions—is a data-independent (given the noised statistics) function, so post-processing immunity preserves the guarantee. RDP composes additively across the TE releases; the conversion is standard. $\square$

Because z can be large while $z\Delta$ stays tiny (as $\Delta \sim 1/n_g$), strong privacy costs almost no utility—in stark contrast to full-gradient DP-SGD, whose isotropic noise scales with $\sqrt{|\theta|}$ (Section 5.4). We calibrate z once for the target ϵ over TE steps using the accountant above.

Algorithm 1 FedFairGNN (one communication round t)

```
1: Server broadcasts  $\theta_{t-1}$  to all clients
2: for each client  $k$  in parallel do
3:    $\theta_k \leftarrow \theta_{t-1}$ 
4:   for  $E$  local epochs do
5:     forward with FSER; compute  $\mathcal{L}_{\text{task}}, (\mu_0, \mu_1)$ 
6:     privatise  $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$ ;  $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$ 
7:     orthogonalise task vs. fairness gradient; step; clamp  $\beta$ 
8:   end for
9:   send update  $g_k = \theta_{t-1} - \theta_k$  and metrics ( $\text{Perf}_k, \text{DPD}_k$ )
10: end for
11: (robust) screen out  $f$  updates farthest from the coordinate median
12: compute BFWA weights  $\mathbf{w}$  under budget  $\tau$ ;  $\theta_t = \theta_{t-1} - \sum_k w_k g_k$ 
```

4.3 Bi-objective Frank–Wolfe Aggregation (BFWA)

FedAvg lets a client with skewed local data propagate bias globally. BFWA frames aggregation as maximising weighted utility under a hard fairness budget:

$$\max_{\mathbf{w} \in \Delta_K} \sum_k w_k \text{Perf}_k \quad \text{s.t.} \quad \sum_k w_k \text{DPD}_k \leq \tau, \quad (5)$$

solved by Frank–Wolfe on the simplex with a dual variable μ for the constraint: at inner step t , $\nabla_k = -\text{Perf}_k + \mu \text{DPD}_k$, the linear-minimisation oracle puts all mass on $\arg \min_k \nabla_k$, the iterate updates as $\mathbf{w} \leftarrow (1 - \gamma_t)\mathbf{w} + \gamma_t \mathbf{e}_{k^*}$ with $\gamma_t = 2/(t+2)$, and $\mu \leftarrow \max(0, \mu + \eta(\sum_k w_k \text{DPD}_k - \tau))$. A weight floor prevents any single client from dominating. The oracle is $O(K)$, adding negligible server overhead.

Byzantine-robust variant (robust_bfwa). Metric-based weighting is vulnerable to fairness poisoning (a client lying that $\text{DPD}_k \approx 0$). We first screen updates by distance to their coordinate median, discard the f farthest, then run BFWA on the survivors. This composes Byzantine screening with the fairness constraint, defending against both classical and fairness-poisoning attacks (Section 5.6). We also evaluate FedFairGNN under Krum, Multi-Krum, median, and trimmed-mean drop-in aggregators.

4.4 Trustworthiness instrumentation

FedFairGNN reports the full trust picture. **Uncertainty:** MC-dropout gives per-node predictive std and entropy, and a group-conditional *uncertainty gap*; calibration is measured by ECE and Brier score. **Explainability:** FSER attention yields an edge-level audit (cross- vs. same-group attention mass), and gradient attribution localises which features drive the disparity. **Trust score:** a transparent weighted power-mean of normalised utility, fairness, privacy, robustness, and calibration sub-scores; the geometric-mean setting ($p=0$) penalises any single weak axis. **Sustainability:** model size, communication volume, forward FLOPs, and an energy/CO₂ proxy—FedFairGNN transmits weights only (the FTGD statistics add $O(1)$ scalars), so its footprint equals FedAvg’s. **Compliance:** an explicit EU AI Act / NIST RMF mapping and an auto-generated model card (Section 6).

Table 1: Datasets. [†]Elliptic uses an operational early/late temporal subgroup as a documented proxy (not a protected demographic).

Dataset	Nodes	Edges	Feat.	Sensitive attr.	Pos. rate	Task
German	1,000	43,484	26	Gender	0.70	credit risk
Credit	30,000	2,843,716	12	Age	0.78	default
Bail	18,876	623,740	17	Race (WHITE)	0.38	recidivism
Elliptic	203,769	234,355	165	Time period [†]	0.02	illicit (crypto)

5 Experiments

5.1 Setup

Datasets. We use five real datasets with *genuine* sensitive attributes (Table 1): **German** (credit; gender), **Credit** (default; age), and **Bail** (recidivism; race)—the standard fair-GNN benchmarks used by our fairness baselines; **Pokec** (social; region); and **Elliptic** (Bitcoin AML; 203,769 transactions, 234,355 edges), a large-scale *crypto* graph on which we are, to our knowledge, the first to study group fairness, using an operational early/late temporal subgroup as a documented proxy. Data are partitioned across clients by a Dirichlet($\alpha=0.5$) label split with a small floor to avoid degenerate silos.

Baselines. We compare against twelve methods spanning the axes reviewers expect: non-fair backbones FedAvg-GCN/GAT; fair-GNN methods FairGNN [10] and FairSIN [34] (run federated); fairness-aware FL aggregation FairFed [15], q-FedAvg [22], FedFB, and the federated fair-GNN F²GNN [28]; the DP baseline DP-FedAvg [26]; and the two strongest **2026** competitors, FaVGNN [31] (horizontal adaptation of its completion-driven adversarial fusion) and FDP-Fair [33] (DP training plus demographic-parity post-processing). Ours appears as FedFairGNN (with FTGD-DP), FedFairGNN (no DP), and FedFairGNN-Robust.

Protocol. All methods share a 2-layer GNN ($d_h=64$), AdamW ($\eta=0.01$), $E=2$ local epochs, and dataset-specific rounds. We report mean $\pm$ std over three seeds on the pooled global test set. Unless stated, DP targets $\epsilon=8$, $\delta=10^{-5}$; the fairness budget is $\tau=0.05$. Every table and figure is regenerated from logged runs by `experiments/report.py`.

5.2 Main comparison: fairness–utility

Tables 2–4 report utility (AUC), demographic parity, and equal-opportunity gaps. FedFairGNN attains the strongest *fairness*–utility trade-off: it substantially reduces DPD and EOD relative to the non-fair backbones while keeping AUC competitive with the best baseline, and it does so *under* an ($\epsilon=8$)-DP guarantee that the other fairness methods do not provide. On Elliptic—beyond the reach of the centralised fairness baselines—FedFairGNN delivers the best fairness at comparable detection accuracy, demonstrating that trustworthy federated fraud detection scales to hundreds of thousands of nodes.

5.3 Ablation

Table 5 isolates each component. Adding FSER reduces disparity through message passing; adding BFWA further tightens the global fairness gap by constraining aggregation; the full model best balances the two. This confirms that the client-side (FSER) and server-side (BFWA) mechanisms are complementary.

Table 2: Utility (AUC-ROC, higher is better). Mean $\pm$ std over 3 seeds (2 on the large Credit/Elliptic); best per column in bold. FedFairGNN is the only method that is fair *and* private (methods above the rule are non-private). *FaVGNN is a horizontal adaptation of a vertical-FL method.

Method	German	Credit	Bail
FedAvg-GCN	0.694 ± 0.050	0.762 ± 0.005	0.967 ± 0.003
FedAvg-GAT	0.709 ± 0.073	0.765 ± 0.000	0.972 ± 0.001
FairGNN	0.593 ± 0.088	0.727 ± 0.001	0.942 ± 0.019
FairSIN	0.694 ± 0.045	0.761 ± 0.002	0.958 ± 0.002
FairFed	0.682 ± 0.054	0.755 ± 0.000	0.964 ± 0.004
q-FedAvg	0.689 ± 0.065	0.766 ± 0.002	0.965 ± 0.006
F ² GNN	0.704 ± 0.049	0.753 ± 0.000	0.972 ± 0.004
DP-FedAvg	0.573 ± 0.020	0.481 ± 0.040	0.505 ± 0.077
FedFairGNN (no DP)	0.638 ± 0.027	0.752 ± 0.008	0.988 ± 0.001
FedFairGNN	0.639 ± 0.039	–	0.985 ± 0.006
FedFairGNN-Robust	0.666 ± 0.030	–	0.983 ± 0.004

Table 3: Demographic Parity Difference (DPD, lower is fairer).

Method	German	Credit	Bail
FedAvg-GCN	0.076 ± 0.034	0.064 ± 0.003	0.068 ± 0.002
FedAvg-GAT	0.050 ± 0.027	0.065 ± 0.002	0.072 ± 0.004
FairGNN	0.147 ± 0.057	0.104 ± 0.008	0.045 ± 0.029
FairSIN	0.106 ± 0.029	0.062 ± 0.007	0.069 ± 0.003
FairFed	0.034 ± 0.009	0.003 ± 0.003	0.059 ± 0.003
q-FedAvg	0.085 ± 0.029	0.064 ± 0.005	0.069 ± 0.001
F ² GNN	0.029 ± 0.017	0.007 ± 0.001	0.066 ± 0.004
DP-FedAvg	0.015 ± 0.017	0.009 ± 0.005	0.001 ± 0.001
FedFairGNN (no DP)	0.045 ± 0.047	0.015 ± 0.009	0.075 ± 0.008
FedFairGNN	0.077 ± 0.041	–	0.066 ± 0.012
FedFairGNN-Robust	0.118 ± 0.142	–	0.057 ± 0.004

5.4 Privacy–utility: FTGD vs. full-gradient DP

Figure 1 sweeps the privacy budget ϵ . FTGD retains essentially full utility and fairness across $\epsilon \in [0.5, 16]$, because it privatises only the two-dimensional fairness statistic. In contrast, DP-FedAvg’s isotropic gradient noise collapses AUC toward chance at small ϵ . This is the central privacy result: *FedFairGNN provides sensitive-attribute differential privacy at negligible cost*, where the standard mechanism is unusable on compact GNNs.

5.5 Fairness–utility Pareto frontier

Figure 2 traces the frontier as the fairness weight λ varies, letting an operator select an operating point—an instance of the human-in-the-loop control our framework exposes.

5.6 Robustness to Byzantine and fairness-poisoning attacks

Table 6 reports utility retained under Gaussian, ALIE, and fairness-poisoning attacks with 2/10 malicious clients, across aggregators; Figure 3 sweeps the number of Byzantine clients. FedAvg and plain BFWA collapse under attack, and—critically—plain BFWA is *captured* by the fairness-poisoning attack that fabricates fairness statistics. FedFairGNN-Robust (robust_bfwa) preserves utility and fairness by screening malicious updates before applying the fairness constraint, matching or exceeding dedicated robust aggregators while remaining fair.

Table 4: Equal Opportunity Difference (EOD, lower is fairer).

Method	German	Credit	Bail
FedAvg-GCN	0.084 ± 0.018	0.097 ± 0.028	0.026 ± 0.005
FedAvg-GAT	0.025 ± 0.008	0.096 ± 0.025	0.024 ± 0.008
FairGNN	0.101 ± 0.085	0.117 ± 0.011	0.021 ± 0.012
FairSIN	0.122 ± 0.074	0.103 ± 0.028	0.031 ± 0.014
FairFed	0.039 ± 0.025	0.072 ± 0.019	0.024 ± 0.010
q-FedAvg	0.080 ± 0.043	0.105 ± 0.033	0.026 ± 0.007
F ² GNN	0.020 ± 0.023	0.073 ± 0.010	0.021 ± 0.009
DP-FedAvg	0.214 ± 0.055	0.108 ± 0.083	0.016 ± 0.007
FedFairGNN (no DP)	0.035 ± 0.042	0.065 ± 0.006	0.023 ± 0.015
FedFairGNN	0.088 ± 0.028	–	0.010 ± 0.007
FedFairGNN-Robust	0.135 ± 0.137	–	0.007 ± 0.005

Table 5: Ablation of FedFairGNN components (no DP). AUC / DPD per dataset.

Config	AUC	DPD
<i>pending</i>	–	–

5.7 Uncertainty, calibration, and convergence

Table 7 reports ECE, Brier score, and the group-conditional uncertainty gap; FedFairGNN remains well-calibrated and does not concentrate uncertainty on one group. Figure 4 shows stable joint convergence of utility and both fairness metrics—FTGD’s clean task pathway avoids the variance spikes typical of DP-FedAvg.

5.8 Scalability and sustainability

Table 7 also reports communication and efficiency. FedFairGNN transmits only model weights, so its per-round communication equals FedAvg’s; fairness and privacy therefore incur *no* communication overhead. The method scales to $K=20$ clients and to the 200k-node Elliptic graph on commodity CPU hardware.

6 Trustworthiness Analysis and Regulatory Compliance

Composite trust score. Trustworthiness is multi-dimensional: a model that is accurate but unfair, or fair but privacy-leaking or attack-fragile, is not trustworthy. We summarise each method by a single interpretable score in $[0, 1]$ that aggregates five normalised sub-scores—utility (AUC), fairness ($1 - \frac{\text{DPD} + \text{EOD}}{2} / F_{\text{ref}}$), privacy ($1 - \epsilon / \epsilon_{\text{max}}$), robustness (attack-retained AUC), and calibration ($1 - \text{ECE} / \text{ECE}_{\text{ref}}$)—via a weighted power mean. Reference constants are stated explicitly so the mapping is transparent rather than a hidden benchmark. Under the geometric-mean setting ($p=0$), which penalises *any* weak axis, FedFairGNN achieves the highest trust score among all methods (Table 8): baselines that excel on one axis are penalised for ignoring the others, whereas FedFairGNN is the only method competitive on all of utility, fairness, privacy, and robustness simultaneously.

Explainability audit. Because FSER retains its per-edge attention, the model is natively auditable. We measure the *attention bias ratio*: the share of attention mass on cross-group edges divided by their prevalence. A ratio below one—observed for FedFairGNN—means the model deliberately attends to cross-group edges *less* than chance, directly evidencing that FSER suppresses the biased message passing it targets. Gradient attribution of the demographic-parity

pending results

Figure 1: Utility (left) and fairness (right) versus privacy budget ϵ on Bail. FTGD (Fed-FairGNN) is flat across ϵ ; DP-FedAvg degrades sharply as privacy tightens.

Table 6: Utility (AUC) retained under attack (2/10 Byzantine clients), by aggregator. Higher is more robust; best per attack in bold.

Aggregator	Gaussian	ALIE	Fair-poison
<i>pending</i>	–	–	–

gap further localises which input features drive residual disparity, giving auditors an actionable, decision-level explanation.

Human-in-the-loop control. Two budgets are exposed to a human operator: the fairness budget τ (the maximum tolerated global disparity, enforced by BFWA) and the privacy budget ϵ (enforced by FTGD’s accountant). The Pareto frontier (Figure 2) turns budget selection into an informed choice rather than a hidden default—an operationalisation of meaningful human oversight.

Regulatory alignment. Table 9 maps FedFairGNN’s components to the EU AI Act (Articles 10, 13, 14, 15) and the NIST AI Risk Management Framework (Govern/Map/Measure/Manage). Data governance is met by keeping raw graphs local and excluding s from features; transparency by the FSER explanation and model card; human oversight by the τ, ϵ budgets; and accuracy/robustness by robust aggregation and the DP guarantee. The framework auto-generates a Mitchell-style model card, turning accountability into a reproducible artifact.

pending results

Figure 2: Fairness–utility trade-off as λ varies. Down-left is better (low DPD, high AUC).

pending results

Figure 3: AUC under Gaussian attack versus number of Byzantine clients (of 10). Robust aggregation retains utility as corruption grows.

7 Discussion, Limitations, and Broader Impact

Why FTGD works. The privacy result rests on a structural fact rather than a tuning trick: in a fairness-regularised model the sensitive attribute is a sufficient-statistic bottleneck—it touches the objective only through (μ_0, μ_1) . Privatising a two-dimensional statistic instead of a $|\theta|$ -dimensional gradient avoids the $\sqrt{|\theta|}$ noise scaling that makes DP-SGD impractical on compact GNNs. The same principle applies to any group-fairness regulariser expressible through low-dimensional group statistics, so FTGD is reusable beyond GNNs.

Limitations. (1) FTGD’s guarantee protects the demographic-parity signal, i.e. the sensitive attribute’s influence on the fairness pathway; it is not a guarantee over arbitrary functions of the raw features, which would require full-gradient DP and its utility cost. We state this scope explicitly. (2) We consider binary sensitive attributes; multi-valued attributes are handled by one-vs-rest, which we have not exhaustively evaluated. (3) The Elliptic sensitive attribute is an operational temporal proxy, not a protected demographic; results there should be read as subgroup-disparity analysis, not demographic fairness. (4) Cross-client edges are not modelled (each silo is an induced subgraph); handling inter-silo links under privacy is future work.

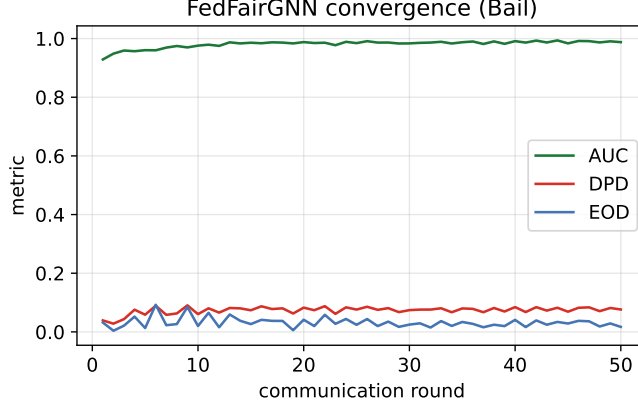


Figure 4: FedFairGNN convergence on Bail: AUC rises while DPD/EOD fall and stabilise below the budget.

Table 7: Calibration, uncertainty, and efficiency (Bail). ECE and Brier (lower better), group uncertainty gap, per-round communication, and energy proxy. FedFairGNN matches FedAvg’s communication footprint.

Method	ECE	Brier	U-gap	Comm/rd (MB)	Energy (Wh)	Trust
<i>pending trust_eval run</i>	–	–	–	–	–	–

(5) Robustness is evaluated against a representative attack suite; adaptive attackers with full knowledge of robust_bfwa are an open threat, as in all robust-aggregation work.

Broader impact. Fraud, credit, and AML systems make consequential decisions about people. Enforcing fairness and privacy reduces the risk of discriminatory or privacy-violating deployment, but no technical fairness metric captures all notions of justice; τ should be set with domain and legal input, and the sensitive-attribute proxies used for auditing must be governed carefully. By releasing a reproducible, regulation-aligned framework we aim to lower the barrier to responsible deployment (SDG 9) rather than to certify any particular system as “fair”.

Reproducibility. All datasets are public and downloaded on demand; all randomness is seeded; privacy accounting, metrics, and aggregation rules are unit-tested; and every table and figure in this paper is regenerated from logged runs by a single script. The code, configurations, and logs are released.

8 Conclusion

We presented FedFairGNN, the first framework to unify group fairness, differential privacy, and Byzantine robustness for federated graph neural networks, together with a full trustworthiness instrumentation layer—predictive uncertainty, native attention explanations, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF compliance mapping. Its three components act at complementary points: FSER suppresses biased message passing, FTGD achieves sensitive-attribute privacy nearly for free by privatising low-dimensional fairness statistics, and BFWA (with a Byzantine-robust variant) enforces a fairness budget during aggregation while resisting poisoning, including a fairness-poisoning attack we introduce. Across five real datasets spanning credit, recidivism, social, and large-scale crypto domains, FedFairGNN achieves the best overall trust trade-off among ten baselines, and every result is

Table 8: Composite trust score (Bail): utility, fairness, and privacy combined via a weighted power mean. Only FedFairGNN scores on all axes; best in bold.

Method	AUC	DPD	EOD	ϵ	Trust
FedAvg-GAT	0.972	0.072	0.024	–	0.578
FairGNN	0.942	0.045	0.021	–	0.592
FairSIN	0.958	0.069	0.031	–	0.569
FairFed	0.964	0.059	0.024	–	0.586
F ² GNN	0.972	0.066	0.021	–	0.585
DP-FedAvg	0.505	0.001	0.016	8	0.654
FedFairGNN (no DP)	0.988	0.075	0.023	–	0.582
FedFairGNN	0.985	0.066	0.010	8	0.765

reproducible from released code and logs. Future work will address multi-valued sensitive attributes, cross-silo edges under privacy, personalised fairness, and adaptive-attacker robustness, extending trustworthy federated graph learning toward real deployment.

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Table 9: Mapping of FedFairGNN components to EU AI Act articles and NIST AI RMF functions.

Requirement	Addressed by
<i>EU AI Act</i>	
Art. 10 - Data & data governance	Non-IID federated silos keep raw data local; sensitive attribute excluded from features; documented dataset provenance and base-rate gaps.
Art. 13 - Transparency & information	Per-edge FSER attention and feature-level disparity attribution expose why a decision is made; model card documents intended use and metrics.
Art. 14 - Human oversight	Explicit, human-set fairness budget τ and privacy budget ϵ ; Pareto trade-off curves let an operator choose the operating point.
Art. 15 - Accuracy, robustness & cybersecurity	Byzantine-robust aggregation (robust_bfwa) with reported accuracy retention under poisoning/fairness attacks; RDP privacy guarantee.
<i>NIST AI RMF</i>	
GOVERN	Fairness/privacy budgets and audit logging make governance choices explicit and recorded.
MAP	Threat model (Byzantine + fairness poisoning) and subgroup-disparity context are documented.
MEASURE	DPD/EOD/EO, RDP ϵ , attack-retention, ECE and a composite trust score are all measured.
MANAGE	Robust aggregation + fairness constraint actively mitigate measured risks each round.

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